Home Work 1

Ex. 0.2.4 Sauer

a) 11.25

Integer Part: $11 = 1011_2$

Decimal $= .25$

$0.25 \times 2 = 0.5$ 0

$0.5 \times 2 = 1.0$ 1

$\Rightarrow 0.01_2$

$\Rightarrow \boxed{1011.01_2}$

b) $\frac{2}{3}$

$\frac{2}{3} \times 2 = \frac{4}{3}$ 1

$\frac{1}{3} \times 2 = \frac{2}{3}$ 0

$\frac{2}{3} \times 2 = \frac{4}{3}$ 1

$\frac{1}{3} \times 2 = \frac{2}{3}$ 0

$\Rightarrow \boxed{0.101010\ldots_2}$

c) $\frac{3}{5}$

$3/5 \times 2 = 6/5$ 1

$1/5 \times 2 = 2/5$ 0

$2/5 \times 2 = 4/5$ 0

$4/5 \times 2 = 8/5$ 1

$3/5 \times 2 = 6/5$ 1

$1/5 \times 2 = 2/5$ 0

$2/5 \times 2 = 4/5$ 0

$4/5 \times 2 = 8/5$ 1

$3/5 \times 2 = 6/5$ 1

$\Rightarrow \boxed{0.100110011\ldots_2}$

d) 3.2

Integer $3 = 11_2$

$$0.2 \times 2 = 0.4 \quad 0$$
$$0.4 \times 2 = 0.8 \quad 0$$
$$0.8 \times 2 = 1.6 \quad 1$$
$$0.6 \times 2 = 1.2 \quad 1$$
$$0.2 \times 2 = 0.4 \quad 0$$
$$0.4 \times 2 = 0.8 \quad 0$$

$0.2 = 0.001100\ldots_2$

$\Rightarrow \boxed{11.001100\ldots_2}$

e) 30.6

$30 = 11110_2$

$$0.6 \times 2 = 1.2 \quad 1$$
$$0.2 \times 2 = 0.4 \quad 0$$
$$0.4 \times 2 = 0.8 \quad 0$$
$$0.8 \times 2 = 1.6 \quad 1$$
$$0.6 \times 2 = 1.2 \quad 1$$
$$0.2 \times 2 = 0.4 \quad 0$$
$$0.4 \times 2 = 0.8 \quad 0$$
$$0.8 \times 2 = 1.6 \quad 1$$

$0.6 = 0.10011001\ldots_2$

$\Rightarrow 11110.10011001\ldots_2$

f) 99.9

$99 = 1100011_2$

$$0.9 \times 2 = 1.8 \quad 1$$
$$0.8 \times 2 = 1.6 \quad 1$$
$$0.6 \times 2 = 1.2 \quad 1$$
$$0.2 \times 2 = 0.4 \quad 0$$
$$0.4 \times 2 = 0.8 \quad 0$$
$$0.8 \times 2 = 1.6 \quad 1$$
$$0.6 \times 2 = 1.2 \quad 1$$

$0.9 = 1110011\ldots_2$

$\Rightarrow \boxed{1100011.1110011\ldots_2}$

EX. 0.2.8, Saver

a) 11011_2 $\quad 1(2^4) + 1(2^3) + 0(2^2) + 1(2^1) + 1(2^0)$

$\Rightarrow \boxed{27_{10}}$

b) 110111.001_2 $\quad 1(2^5) + 1(2^4) + 0(2^3) + 1(2^2) + 1(2^1) + 1(2^0)$

$\Rightarrow 55$

$0(2^{-1}) + 0(2^{-2}) + 1(2^{-3})$

$\Rightarrow 0.125$

$\boxed{55.125_{10}}$

c) $111.\overline{001}_2$ $\quad 1(2^2) + 1(2^1) + 1(2^0)$

7

$2^3 = 8$

$8x = 1.001\ldots_2$

$8x - x = 1$

$7x = 1$

$x = \frac{1}{7}$

$7 + \frac{1}{7} = \frac{50}{7} \approx \boxed{7.14285\ldots_{10}}$

d) $1010.\overline{01}_2$ $\quad 1010 = 10$ $\quad 01 \; 01 \; 01 \; 01 \ldots$

$\frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^6} + \ldots = \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \ldots$

Geometric Sereies $\quad a = \frac{1}{4}, \; r = \frac{1}{4}$

$$\frac{a}{1-r} = \frac{1/4}{1-1/4} = \frac{1/4}{3/4} = \frac{1}{3}$$

$10 + \frac{1}{3} = \frac{31}{3} \approx \boxed{10.333_{10}}$

e)

10111.10101 $\quad 10111 = 23_{10}$

$1(2^{-1}) + 0(2^{-2}) + 1(2^{-3}) + 0(2^{-4}) + 1(2^{-5})$

$\Rightarrow .65625_{10}$

$\boxed{23.65625_{10}}$

f)

$1111.01\overline{001}$ $\quad 1111_2 = 15_{10}$

$0(2^{-1}) + 1(2^{-2}) + 0(2^{-3})$

$0.25 = 1/4$

$\frac{1}{2^6} + \frac{1}{2^9} + \frac{1}{2^{12}} + \ldots = \frac{1}{64} + \frac{1}{512} + \ldots$

Geometric series $a = \frac{1}{64}$ $\quad r = \frac{1}{8}$ $\quad \frac{1/64}{7/8} = \frac{1}{64} \cdot \frac{8}{7} = \frac{1}{56}$

$\frac{1}{4} + \frac{1}{56} = \frac{15}{56}$

$15 + \frac{15}{56} = \frac{855}{56} \approx 15.2678_{10}$

EX 0.3.6, Sauer

a) $(1 + (2^{-51} + 2^{-52} + 2^{-54})) - 1$

$x = 2^{-51} + 2^{-52} + 2^{-54}$

$1 + s = 1.\underbrace{00\ldots}_{50 \text{ zeros}}011010\ldots$

bits at position 51 52 54 are 1 everything eles is 0

52nd bit is 2^{-52} anything at 2^{-53} or smaller is rounded

$$Fl(1+x) = 1.\underbrace{00\cdots0}_{50 \text{ zeros}}11$$

$$\Rightarrow 1 + 2^{-51} + 2^{-52}$$

$$\Rightarrow Fl(1+s) = 2^{-51} + 2^{-52}$$

$$\Rightarrow \boxed{3 \times 2^{-52}}$$

b)

$$(1 + (2^{-51} + 2^{-52} + 2^{-60})) - 1$$

$$x = 2^{-51} + 2^{-52} + 2^{-60}$$

$$1 + x = 1.\underbrace{00\cdots0}_{50 \text{ zeros}}11\underbrace{00\cdots0}_{7 \text{ zeros}}1\bar{0}\cdots$$

$$Fl(1+x) = 1.\underbrace{00\cdots0}_{50}11 = 1 + 2^{-51} + 2^{-52}$$

$$Fl(1+x) - 1 = 2^{-51} + 2^{-52}$$

$$\Rightarrow \boxed{3 \times 2^{-52}}$$

EX 0.3.7 Langou

a) 16 $16 = 2^4 \times 1.0$

sign = 0

Exponent: $4 + 1023 = 1027 = 100\ 0000\ 0011_2$

mantissa: all zeros

Bits: $\underbrace{0\ 100}_{4}\underbrace{0000}_{0}\underbrace{0011}_{3}$ 0000...0

$\Rightarrow \boxed{4030000000000000}$

b) 130

$130 = 2^7 \times 1.0000010_2$

sign: 0

Exponent: $7 + 1023 = 1030 \rightarrow 100\ 0000\ 0110$

mantissa: bit 6 = 1, rest are 0

Bits: 0 10000000110 000001 0000 ... 0

$\Rightarrow \boxed{4060400000000000}$

c) 1/4 $\quad 1/4 = 2^{-2} \times 1.0$

Sign: 0

Exponent: $-2 + 1023 = 1021 \Rightarrow 011\ 1111\ 1101$

Mantissa: all 0's

Bits: 0 01111111101 0000...0

$\boxed{3FD\,0000000000000}$

d) fl(1/7)

$1/7 = 2^{-3} \times 1.\overline{001}_2$

Sign: 0

Exponent: $-3 + 1023 = 1020 \rightarrow 011\ 1111\ 1100$

Mantissa: 001 001 001 001 001 001 001 001 001 001 001 001 001 001 001 001 10

0010	0100	1001	0010	0100	1001	0010	0100	1001	0010	0100	1001	0010
2	4	9	2	4	9	2	4	9	2	4	9	2

0 01111111100 001001...

$\boxed{3FC\,2492492492\,49}$

e)

$fl(4/7)$ $\qquad 4/7 = 2^{-1} \times (8/7)$ and $8/7 = 1 + 1/7 = 1.\overline{001}_2$

mantisa = 001 repeating for 52 bits

Fraction bits:
(52 bits)

0010 0100 1001 0010 0100 1001 0010 0100 1001 0010 0100 1001 0010

53^{rd} bit: 0

Sign: 0

Exponent: $-1 + 1023 = 1022 \Rightarrow$ 011 1111 1110

$\boxed{3FE2492492492492}$

f)

$fl(0.01)$

0.01 x 2	=	0.02	0
0.02 x 2	=	0.04	0
0.04 x 2	=	0.08	0
0.08 x 2	=	0.16	0
0.16 x 2	=	0.32	0
0.32 x 2	=	0.64	0
0.64 x 2	=	1.28	1
0.28 x 2	=	0.56	0
0.56 x 2	=	1.12	1
0.12 x 2	=	0.24	0
0.24 x 2	=	0.48	0
0.48 x 2	=	0.96	0
0.96 x 2	=	1.92	1

repeating block
00011101000111101 0111

0.01 =

$0.00000001010001111010111\overline{00011101000111101 0111}$

Normalize: $1.01000111101011100001 \times 2^{-7}$

Mantisa 52 bits

Block 1 (1-20): 01000111101011100001

Block 2 (21-40): 01000111101011100001

Block 3 (41-52): 010001111010

Bit 53: 1 next bit in Pattern 1

Sign: 0

Exponent: $-7 + 1023 = 1016 \Rightarrow 011\ 1111\ 1000$

Mantissa:

0100	0111	1010	1110	0001	0100	0111	1010	1110	0001	0100	0111	1011
4	7	A	E	1	4	7	A	E	1	4	7	B

$\boxed{3F847AE147AE147B}$

g) fl(-0.01

same a f but flip the sign bit

1011	1111	1000	0100	0111	1010	1110	0001	0100	0111	1010	1110	0001	0100	0111	1011
B	F	8	4	7	A	E	1	4	7	A	E	1	4	7	B

$\boxed{BF847AE147AE147B}$

h) fl(-0.02)

$-0.02 = -2 \times 0.01 = -1.01000111101011100001 \times 2^{-6}$

Mantisa: same as f

Sign: 1

Exponent: $-6 + 1023 = 1017 \Rightarrow 011\ 1111\ 1001$

1011	1111	1001	0100	0111	1010	1110	0001	0100	0111	1010	1110	0001	0100	0111	1011
B	F	9	4	7	A	E	1	4	7	A	E	1	4	7	B

$\boxed{BF947AE147AE147B}$

EX 0.4.2 Saver

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x^2 + 3x - 8 = 0$$

$$x = \frac{-3 \pm \sqrt{9 - 4(-8^{-14})}}{2}$$

$$\Rightarrow \quad \Delta = \sqrt{9 + 4 \cdot 2^{-42}}$$

$$\Rightarrow \quad \Delta \; 3\sqrt{1 + \frac{2^{-40}}{9}} \approx 3\left(1 + \frac{2^{-41}}{9}\right) = 3 + \frac{2^{-41}}{3}$$

$$\Rightarrow \quad x_1 = \frac{-3 - \Delta}{2} \approx \frac{-3 - 3}{2} = -3$$

$$\boxed{x_1 = -3.00}$$

$$\Rightarrow \quad x_1 \cdot x_2 = -8^{-14} = -2^{-42}$$

$$\Rightarrow \quad x_2 = \frac{-2^{-42}}{x_1} = \frac{-2^{-42}}{-3 - \frac{2^{-41}}{6}} \approx \frac{2^{-42}}{3} = \frac{1}{3} \cdot 2^{-42}$$

$$\Rightarrow \quad 2^{-42} \approx 2.274 \times 10^{-13}$$

$$x_2 = \frac{2.274 \times 10^{-13}}{3} \approx \boxed{7.58 \times 10^{-14}}$$

EX 0.5.2 Saver

$$f'(c) = \frac{f(1) - f(0)}{1 - 0} = f(1) - f(0)$$

a)

$$f(x) = e^x$$

$$\frac{f(1) - f(0)}{1 - 0} = e^1 - e^0 = e^1 - 1$$

$$\Rightarrow \quad f'(x) = e^x \Rightarrow e^c = e - 1$$

$\Rightarrow \quad c = \ln(e-1) \approx \boxed{0.5413}$

B)

$f(x) = x^2$

$\dfrac{f(1) - f(0)}{1-0} = 1^2 - 0^2 = 1$

$\Rightarrow f'(x) = 2x \Rightarrow 2c = 1$

$\Rightarrow \boxed{c = \frac{1}{2}}$

c) $f(x) = \dfrac{1}{x+1}$

$\dfrac{f(1) - f(0)}{1} = \frac{1}{2} - \frac{1}{1} = -\frac{1}{2}$

$f'(x) = -\dfrac{1}{(x+1)^2} \Rightarrow \dfrac{1}{(c+1)^2} = -\dfrac{1}{2}$

$\Rightarrow (c+1)^2 = 2$

$\Rightarrow c+1 = \sqrt{2}$

$\Rightarrow \boxed{c = \sqrt{2} - 1 \approx 0.414}$

EX 0.5.6 saver

a) $f(x) = x^{-2}$ $\qquad f(1) = 1$

$f'(x) = 2x^{-3}$ $\qquad f'(1) = -2$

$f''(x) = 6x^{-4}$ $\qquad f''(1) = 6$

$f'''(x) = -24x^{-5}$ $\qquad f'''(1) = -24$

$f''''(x) = 120x^{-6}$ $\qquad f''''(1) = 120$

$$\Rightarrow \quad P_4(x) = \sum_{k=0}^{4} \frac{f^{(k)}(1)}{k!}(x-1)^k$$

$$\Rightarrow \quad \frac{1}{0!} + \frac{-2}{1!}(x-1) + \frac{6}{2!}(x-1)^2 + \frac{-24}{3!}(x-1)^3 + \frac{120}{4!}(x-1)^4$$

$$\Rightarrow \quad \boxed{P_4(x) = 1-2(x-1) + 3(x-1)^2 - 4(x-1)^3 + 5(x-1)^4}$$

B)

$f(0.9)$ and $f(1.1)$

let $h = x-1$

- $x = 0.9 \Rightarrow h = -0.1$
- $x = 1.1 \Rightarrow h = 0.1$

At $x = 0.9$

$$P_4(0.9) = 1-2(-0.1) + 3(-0.1)^2 - 4(-0.1)^3 + 5(-0.1)^4$$

$$\Rightarrow \quad 1 + 0.2 + 0.03 + 0.004 + 0.0005$$

$$\Rightarrow \quad \boxed{P_4(0.9) = 0.8265}$$

At $x = 1.1$

$$P_4(1.1) = 1-2(0.1) + 3(0.1)^2 - 4(0.1)^3 + 5(0.1)^4$$

$$\Rightarrow \quad \boxed{P_4(1.1) = 0.8265}$$

c)

$$R_4(x) = \frac{f^{(5)}(c)}{5!}(x-1)^5$$

$$f^{(5)}(x) = -720x^{-7} \Rightarrow |f^{(5)}(c)| = \frac{720}{|c|^7}$$

$$|R_4(x)| = \frac{720}{5! \cdot |c|^7} |x-1|^5 = \frac{720}{120} \cdot \frac{|x-1|^5}{|c|^7} = \frac{6|x-1|^5}{|c|^7}$$

- For $x = 0.9$: $c \in (0.9, 1)$, so $|c| \geq 0.9$
- For $x = 1.1$: $c \in (1, 1.1)$, so $|c| \geq 1.0$

Error bound at $x = 0.9$:

$$|R_4(0.9)| \leq \frac{6 \cdot (0.1)^5}{(0.9)^7} = \frac{6 \times 10^{-5}}{0.4783} \approx \boxed{1.254 \times 10^{-4}}$$

Error bound at $x = 1.1$:

$$|R_4(1.1)| \leq \frac{6 \cdot (0.1)^5}{(1.0)^7} = 6 \times 10^{-5} = \boxed{6.000 \times 10^{-5}}$$

d)

$$f(0.9) = (0.9)^{-2} = \frac{1}{0.81} = 1.23456790\ldots$$

Error	Bound from c
6.79×10^{-5}	1.254×10^{-4}

$$f(1.1) = (1.1)^{-2} = \frac{1}{1.21} = 0.82644628\ldots$$

Error	Bound from c
5.37×10^{-5}	6.000×10^{-5}